Text-Anchored Deep Optimal Transport Clustering for Remote Sensing Cross-Scene Classification

Yue Zhao, Haoyang Fang, Maoguo Gong, *Fellow, IEEE*, Zhao Wang, *Member, IEEE*, Xin Mei, Hua Yu, Lianzhong Zhang, and Yew-Soon Ong, *Fellow, IEEE*

***Abstract*—Remote sensing scene classification (RSSC) under cross-scene settings aims to transfer semantic knowledge learned from a labeled source domain to an unlabeled target domain that exhibits substantial distribution shifts caused by variations in geography, sensors, and imaging conditions. Existing adaptation methods mainly rely on image-level feature alignment, which leads to weak semantic understanding and consistency across domains and poor generalization in complex remote sensing (RS) scenarios. In this article, we propose a text-anchored deep optimal transport clustering network (TA-DOTCNet) to explicitly introduce category-level textual semantics as external guidance for cross-scene representation learning. The core idea is to leverage domain-invariant textual descriptions to regulate visual adaptation and enforce semantically consistent clustering across heterogeneous domains. First, a text-anchored external guidance strategy (TaGS) integrates textual semantics into visual representation learning through extended prompt construction, cross-scene alignment, and prototype-driven counterpart fusion, enhancing discriminability while preserving semantic coherence across domains. Second, a deep optimal transport clustering (DOTC) module formulates image–text alignment as an entropy-regularized optimal transport (EOT) problem, transforming similarity-based assignments into compact target distributions and enforcing domain-invariant clustering structures. Our proposal effectively mitigates domain bias and improves cross-scene generalization. Extensive experiments on multiple RS benchmarks demonstrate the superiority and robustness of the proposed method over state-of-the-art approaches. The source code is available at https://github.com/bling2beyond/TA-DOTCNet**

***Index Terms*—Cross-scene classification, domain adaptation, multimodal learning, optimal transport (OT), remote sensing (RS).**

I. Introduction

REMOTE sensing scene classification (RSSC) aims to automatically identify the semantic category (e.g., residential, farmland, forest, or industrial area) of land-cover scenes from high-resolution imagery [1], [2], [3], and has become a cornerstone for applications such as land-use mapping [4], urban development [5], and environmental monitoring [6]. With the emergence of deep learning, convolutional and transformer-based networks have greatly advanced RSSC [7], [8], enabling powerful feature representation and robust discrimination of complex ground objects. In recent years, the continuous upgrading of remote sensing (RS) sensors and observation platforms has brought an explosive growth of data with higher spatial, spectral, and temporal resolutions [9], [10], [11]. These advancements allow for finer-scale and multitemporal monitoring of land-cover changes and human activities, presenting unprecedented opportunities for large-scale RS analysis [12], [13]. However, increasing data volume and heterogeneity also raise new challenges. Obtaining high-quality annotated data across diverse scenes, platforms, and acquisition conditions is both expensive and labor-intensive, often requiring domain experts for accurate labeling. As a result, labeled data are typically available only for limited regions, whereas most newly acquired images remain unlabeled.

When a classification model trained on one scene or dataset is directly applied to another scene with different geographic distributions, illumination, or imaging sensors, its performance often deteriorates significantly due to the domain shift problem. This phenomenon reveals that conventional supervised deep networks lack generalization ability across heterogeneous domains. To overcome this limitation, cross-scene classification has been proposed as a promising solution [14], [15]. The goal is to transfer the knowledge learned from labeled source scenes to unlabeled target scenes with distinct but related data distributions. By leveraging domain adaptation and transfer learning strategies, cross-scene classification seeks to reuse the semantic knowledge embedded in the source domain while mitigating the negative influence of distribution discrepancies.

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Most existing methods perform feature distribution alignment or transfer adaptation by leveraging discrepancy-based, class-wise, or subdomain strategies to reduce the distribution gap between source and target domains. For example, Othman et al. [16] proposed a domain adaptation network for cross-scene classification that integrates a pretrained convolutional neural network (CNN) with additional adaptation layers to minimize the maximum mean discrepancy (MMD) and a geometric regularization term between domains. Huang et al. [17] introduced a semi-supervised bidirectional sample-class alignment (BSCA) model that establishes bidirectional sample-to-class consistency to enhance adaptation under limited supervision. Li et al. [18] proposed a spatial-topological-semantic alignment (STSA) framework that jointly regularizes geometric and semantic consistency for improved feature alignment with few source labels. More recently, to further address multisource discrepancies, Wang et al. [19] developed multisource subdomain distribution alignment network (MSSDANet) for cross-domain scene classification, which performs fine-grained subdomain alignment and semantic transfer.

Although these approaches have achieved promising results, they mainly rely on statistical or visual-level alignment and often lack explicit semantic guidance to ensure conceptually consistent adaptation across different scenes. As shown in Fig. 1, it still faces major limitations.

1) Most current cross-scene classification approaches remain confined to the image modality, where adaptation operates solely on visual features. Such image-level alignment is inherently inefficient in information representation, particularly for RSSC that exhibit high interclass similarity and large intraclass variance. As a result, samples of the same land-cover category across different domains often display weak semantic consistency in the feature space, making it difficult to form unified and discriminative representations for cross-scene understanding.
2) Most existing methods either adapt one-way to the target domain or enforce shared-space alignment by pulling the two domains closer. Yet discrepancy minimization alone may overfit to domain proximity by matching domain-specific appearance statistics rather than category-consistent concepts, thus failing to ensure real semantic consistency. As a result, target-domain performance may improve at the expense of reduced discriminability on the source domain.

The same category exhibits different distributions across images in various scenarios (i.e., domain drift), yet they share the same textual semantics. Therefore, introducing the text modality during training effectively guides the accurate extraction of scene semantic information from images, thereby bridging the semantic gap caused by domain drift in the image modality. To this end, we innovatively propose the text-anchored deep optimal transport clustering network (TA-DOTCNet). By incorporating text prompt construction, we enhance the consistency and precision of cross-domain semantic expression. This enables adaptation to the target domain while preserving memory or understanding of the source domain, rather than overfitting to the domain proximity. Specifically, we propose a text-anchored external guidance strategy (TaGS) that leverages textual modality to bridge the semantic gap in distribution across cross-scene image modality. Furthermore, to make cross-scene image modality embeddings converge more toward the text modality, thereby enhancing similar semantic representations shared by the same category across different domains while ensuring more discriminative representations for different categories, we propose a deep optimal transport clustering (DOTC). By leveraging the sparsity of entropy-regularized optimal transport (EOT), this approach enables flexible control over the compactness between image embeddings and text anchors. Experimental results across four datasets demonstrate that TA-DOTCNet significantly enhances cross-scene understanding capabilities.

The contributions of this article can be summarized as follows.

1) We propose a TA-DOTCNet for RS cross-scene classification, which introduces textual semantics as external anchors to guide visual adaptation and bridge the semantic gap across heterogeneous domains caused by domain drift.
2) We design a text-anchored external guidance strategy (TaGS) to align image features with textual concepts for enhanced cross-domain semantic consistency, and further propose DOTC that employs EOT to enforce intraclass compactness and interclass separability in the joint image–text space.
3) Extensive experiments on four benchmark datasets demonstrate that TA-DOTCNet surpasses state-of-the-art cross-scene classification methods, improving cross-domain adaptability while preserving source-domain semantic memory for better generalization and interpretability.

The organization of this article is as follows. First, we briefly introduce the relevant works of cross-scene classification, multimodal learning, deep clustering, and optimal transport (OT) in Section II. In Section III, the proposed TA-DOTCNet, the corresponding component modules will be introduced in detail. Comparisons with advanced methods, ablation analysis, and visualization studies will be introduced in Section IV. Finally, Section V summarizes and looks ahead to future study.

II. Related Works and Background

A. RS Cross-Scene Classification

1) Vanilla RSSC: RSSC aims to automatically assign semantic labels to land-cover scenes from high-resolution imagery. Early studies mainly relied on handcrafted feature descriptors, such as scale-invariant feature transform (SIFT) [20], local binary patterns (LBPs) [4], and histogram of oriented gradients (HOGs) [21], combined with traditional classifiers like support vector machines (SVMs) [22] or random forests [23]. These methods achieved satisfactory performance on small, homogeneous datasets but were limited by their weak representation capability and sensitivity to imaging variations [1]. With the emergence of deep learning, CNNs and transformer-based architectures have become the

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mainstream frameworks for RSSC, significantly improving classification accuracy and robustness [3], [7], [24], [25]. Despite these advances, most existing deep models follow a supervised learning paradigm and rely heavily on large-scale labeled datasets, which are difficult and costly to acquire in practice. Consequently, their generalization ability degrades dramatically when applied to unseen scenes collected under different sensors, seasons, or geographic regions.

2) Advanced Cross-Scene Classification: To alleviate domain shift between heterogeneous scenes, cross-scene classification has become an important research direction. Early studies mainly relied on shallow transfer-learning techniques, such as transfer component analysis (TCA) and geodesic flow kernel (GFK), for hyperspectral and optical scene classification [26], [27], [28]. Later, deep domain adaptation frameworks were introduced to align feature distributions at visual or semantic levels. Representative examples include AMRAN [29], which combines multiscale residual learning with distribution alignment, BSCA [17], which performs BSCA, and the recent source-free method SFSC [30], which further improves robustness through statistics matching and noise adaptation. Although these methods have significantly advanced cross-scene classification, most of them still operate purely in the image modality and rely mainly on visual feature alignment, which remains insufficient for heterogeneous RS data with high interclass similarity and large intraclass variance.

B. Multimodal Learning in RS

In recent years, the success of large-scale vision–language models (VLMs) such as CLIP [31] and BLIP [32] has demonstrated the remarkable capability of textual information to enhance visual understanding. In computer vision, textual descriptions or category names provide high-level semantic cues that help models learn more robust and transferable representations [33], [34]. By aligning visual features with text embeddings in a shared semantic space, these models effectively capture domain-invariant concepts and achieve impressive generalization across unseen datasets or tasks.

Inspired by these advances, the integration of text as an auxiliary or guiding modality has recently been explored in RS. On the one hand, introducing textual semantics can enhance image representation learning by embedding domain knowledge (e.g., land-cover definitions or scene descriptions) into the feature space. This fusion allows RS models to encode higher-level semantics that are less sensitive to imaging variations, thereby improving recognition robustness and interpretability [35], [36]. On the other hand, the text modality can serve as a tool for semantic matching. Most existing works focus on global retrieval or zero-shot recognition [37], [38]. Recently, Zhang et al. [39] further explored prompt-driven semantic guidance for fine-grained building change detection in urban scenarios, showing that prompt-based prior knowledge can enhance RS interpretation under complex domain variations. In this article, we novelly serve text as external supervision information that guides the alignment of features in different domains. Such guidance enables more consistent cross-domain representations, bridging the semantic gap caused by domain drift, and facilitating scene understanding under heterogeneous conditions.

C. Deep Embedded Clustering

Deep embedded clustering (DEC) has become an effective paradigm for jointly learning feature representations and cluster assignments in an end-to-end manner. Unlike conventional clustering methods, e.g., k-means or Gaussian mixture models, that operate on fixed feature spaces, DEC integrates clustering objectives directly into the deep network optimization process. Xie et al. [40] first introduced the DEC framework, which simultaneously learns a nonlinear mapping from the data space to a latent feature space and refines cluster centroids by minimizing the Kullback–Leibler (KL) divergence between soft cluster assignments and their target distribution. Subsequent variants improved cluster stability and discriminability by incorporating reconstruction constraints or contrastive learning strategies. Guo et al. [41] proposed an improved DEC (IDEC), which incorporates an autoencoder-based reconstruction constraint to preserve local data structure and prevent distortion of the feature space during clustering optimization. Wu et al. [42] further proposed deep comprehensive correlation mining (DCCM), which leverages correlation mining and joint learning of multiple clustering objectives to enhance feature discriminability and clustering robustness. This article explores the matching and coupling between the source distribution and the target distribution from the perspective of OT in a multimodal context.

D. OT and Entropy-Regularized Alignment

EOT extends classical OT by introducing an entropic regularizer, which makes the transport plan smoother and computationally tractable. Villani et al. [43] provide the mathematical foundation of OT from the perspective of distribution matching under geometric constraints. Cuturi [44] further showed that entropy regularization enables efficient optimization through Sinkhorn iterations, making OT practical for modern machine learning. Courty et al. [45] demonstrated the effectiveness of OT for domain adaptation, where transport-based alignment can better handle cross-domain distribution shifts. Chen et al. [46] and Zuo et al. [47] further extended OT-based alignment to structured and cross-modal scenarios. Motivated by these works, we build upon the classical EOT formulation and propose to design it as a parameter-free distribution refinement tool to obtain more compact and semantically consistent assignments between image embeddings and textual anchors.

III. Methodology

This section will efficiently present the detailed process and critical components of the TA-DOTCNet, including the text-anchored external guidance strategy (TaGS) and DOTC. Through theoretical proofs and formula derivations, it provides the fundamental theoretical foundation for cross-scenario semantic understanding.

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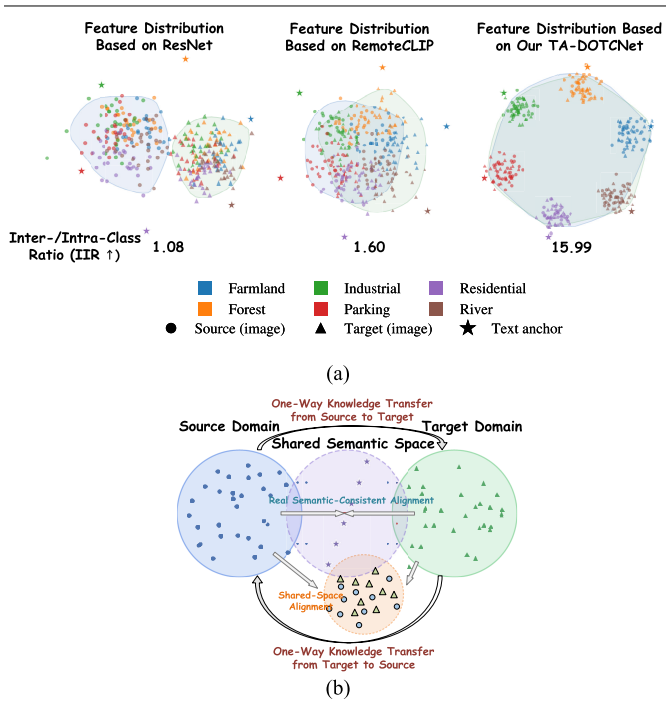


Fig. 1. Motivation of semantic-guided cross-scene representation learning. Feature distributions produced by a ResNet-based classifier, RemoteCLIP, and the proposed TA-DOTCNet are compared. TA-DOTCNet achieves compact, domain-invariant clusters aligned with semantic anchors, demonstrating the advantage of semantic-guided multimodal alignment. (a) Weak discriminability and semantic misalignment in cross-scene understanding. (b) Overfitting to domain proximity over category semantics.

A. Overview of TA-DOTCNet

RS cross-scene classification aims to transfer semantic knowledge learned from a labeled source domain to an unlabeled target domain under substantial domain shifts caused by variations in geography, sensors, seasons, and imaging conditions. Such shifts lead to severe feature drift in the image modality, making conventional image-only alignment strategies insufficient to guarantee semantic consistency across domains. A key observation is that scene-level textual semantics remain invariant across domains, even when visual appearances vary significantly. Motivated by this property, we propose TA-DOTCNet, which introduces category-level textual embeddings as external semantic anchors and formulates cross-scene adaptation as a semantic-guided multimodal alignment and clustering problem.

As illustrated in Fig. 2, TA-DOTCNet adopts a Siamese architecture for source and target domains, where images from both domains are encoded by a shared RemoteCLIP [48] image encoder followed by lightweight cross-scene adapter. In parallel, descriptive text prompts are constructed for each scene category and encoded by a frozen text encoder to form a domain-invariant semantic pool. These textual embeddings provide stable semantic References throughout training. To effectively inject textual semantics into visual representations, TA-DOTCNet introduces a prototype-driven counterpart construction mechanism, which dynamically aggregates category-level textual embeddings according to image prototypes and fuses them with refined image features for supervised classification (See Section III-B). This design enhances discriminability while maintaining semantic coherence across domains.

Beyond feature fusion, TA-DOTCNet further enforces semantic consistency through a parameter-free Deep OT Cluster Head. This module models the alignment between image embeddings and textual anchors as an EOT problem, transforming similarity-based soft assignments into compact target distributions. By minimizing the distributional discrepancy between these assignments via KL divergence, the model explicitly encourages domain-invariant, semantically aligned clustering structures in both image–text and image–image spaces (See Section III-C).

The overall training objective of TA-DOTCNet is defined as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{OTCluster} \tag{1}$$

where $\mathcal{L}_{\text{CE}}$ denotes the supervised cross-entropy loss applied to the multimodal representations, and $\mathcal{L}_{OTCluster}$ is the text-anchored OT clustering loss that aligns image embeddings with category-level textual anchors. λ is a balancing coefficient controlling the contribution of the two losses.

B. Text-Anchored External Guidance Strategy

Cross-scene discrepancies in RS imagery often manifest as substantial variations in illumination, seasonality, sensor conditions, and geographic appearance. These variations cause visual embeddings of the same semantic category to drift apart across domains, making purely image-driven alignment insufficient for achieving stable cross-domain semantic consistency. In contrast, textual descriptions of scene categories remain invariant across acquisition conditions, offering a semantically stable modality that does not suffer from domain shift. Building on this insight, the proposed text-anchored external guidance strategy (TaGS) leverages textual semantics as external anchors to regulate the adaptation of visual representations in both source and target domains.

The collaborative alignment and enhancement between textual and visual information are achieved through three components: text prompt construction, cross-scene alignment and enhancement, and prototype-driven counterpart construction and fusion. While textual descriptions provide preliminary semantic guidance during feature adaptation in TaGS, their anchoring role is mainly realized in the Deep OT Cluster Head (See Section III-C), where text embeddings explicitly guide the finer-grained semantic alignment of image representations.

1) Text Prompt Construction: To obtain a semantically meaningful textual pool, we construct descriptive prompts for each scene category rather than relying on generic templates (GTs) such as "⌊**Class**⌉" or "*This is a photograph of* ⌊**Class**⌉." RS scenes typically encompass characteristic land-cover components, spatial arrangements, and functional attributes; therefore, a richer description yields text embeddings with more discriminative semantic density. Our prompt design principle follows two requirements: rich semantic cues and alignment with RemoteCLIPs textual embedding behavior [48]. Each prompt includes not only the category name but also representative land-cover elements, typical spatial layouts, and visually identifiable structures. Such descriptions

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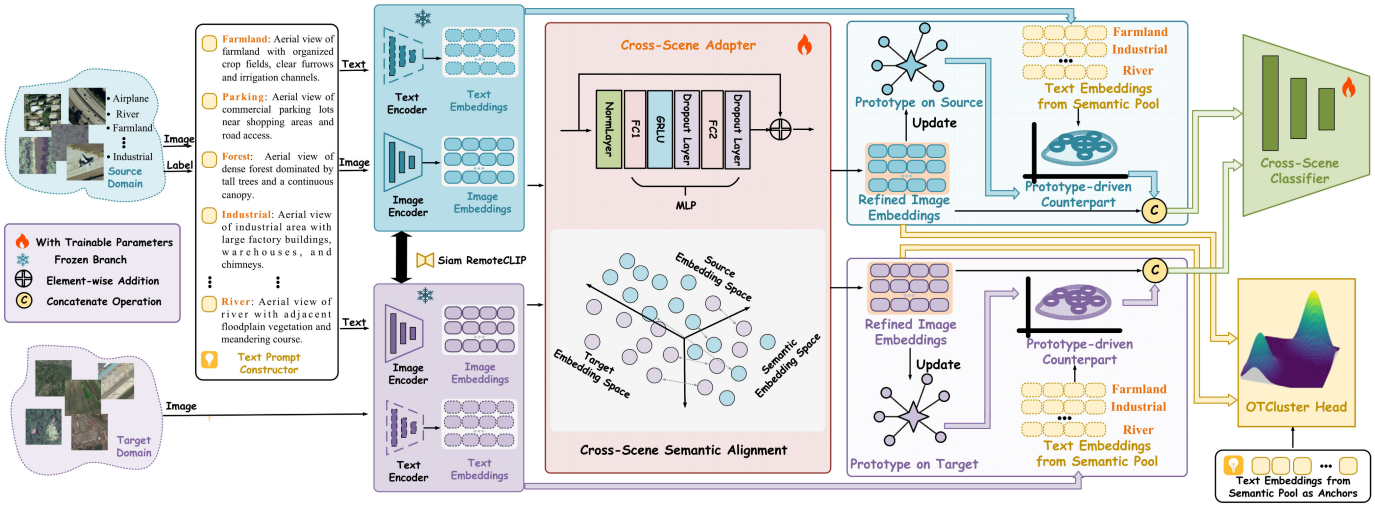


Fig. 2. Overall framework of the proposed TA-DOTCNet for RS cross-scene classification. The network adopts a Siamese architecture for source and target domains, where images are encoded by a shared RemoteCLIP image encoder augmented with cross-scene adaptive layers. Category-level textual prompts are constructed and encoded by a frozen text encoder to form a semantic pool serving as external anchors. Prototype-driven textual counterparts are dynamically generated and fused with refined image embeddings for supervised classification. Meanwhile, a parameter-free Deep OT cluster head leverages text embeddings as anchors to enforce compact and semantically consistent alignment between image representations across domains.

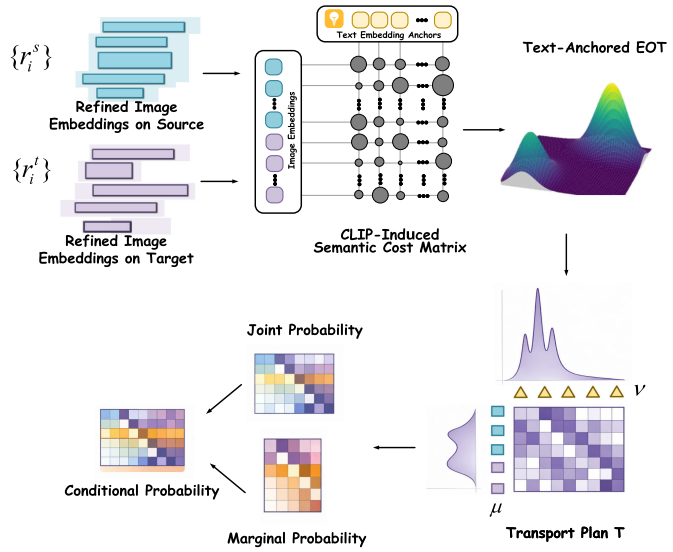


Fig. 3. Overview of the proposed deep OT cluster head. Refined image embeddings and category-level textual anchors are first used to construct a similarity-based source distribution. An EOT formulation refines this distribution into a compact target assignment by converting joint probabilities in the transport plan into conditional distributions, enabling controllable sharpening or smoothing of semantic alignment. The resulting target distribution serves as a parameter-free supervisory signal to enhance semantic consistency across domains.



Fig. 4. Sample images from the four benchmark datasets used in our experiments. (Top to bottom) AID, RSSCN7, UCM, and WHU. These datasets exhibit significant diversity in spatial resolution, scene composition, and environmental conditions, providing a comprehensive benchmark for evaluating cross-scene generalization capabilities.

encode higher-level semantics that remain invariant across domains and better differentiate categories with similar appearance patterns. RemoteCLIP is trained on natural-language descriptions rather than minimal class labels. Using extended descriptions ensures that the resulting text embeddings occupy well-structured regions in the representation space and better align with CLIP-style vision–language contrastive priors. This improves the grounding of textual anchors and enhances cross-modal consistency.

The complete set of textual descriptions used for each category is provided in Table I, where each prompt is generated to reflect the core semantic attributes of its corresponding land-cover scene. These text prompts are then tokenized and encoded using a frozen RemoteCLIP text encoder to obtain fixed semantic embeddings. In particular, the category-level scene descriptions are generated by a large language model (LLM) rather than manually handcrafted for each dataset or image. We query an LLM as:

Provide exactly five single-sentence descriptions for scene "`[Class]`." Each sentence must explicitly contain the category name, mention 2–3 land-cover elements, describe an overhead spatial pattern, include ≥2 visible structures, state an aerial or satellite perspective, stay strictly within 10–25 words, and vary its visual focus.

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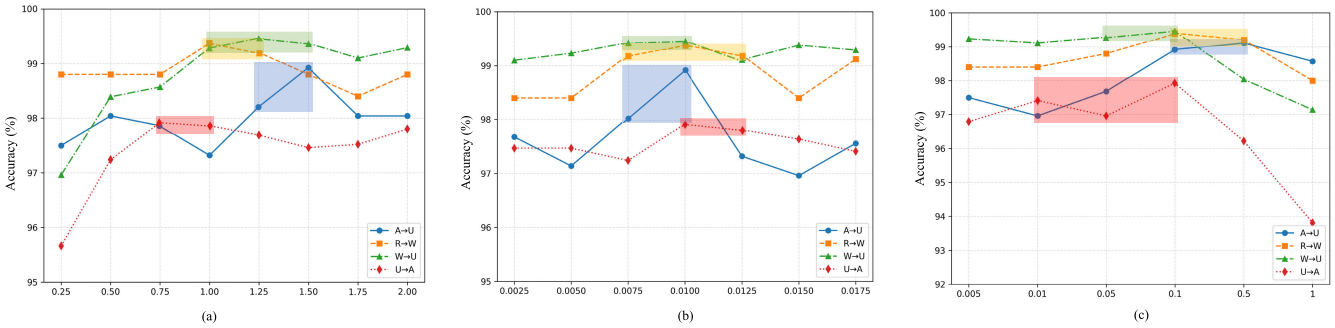


Fig. 5. Sensitivity analysis of TA-DOTCNet with respect to key hyperparameters. Performance variations are reported for different values of the loss balancing coefficients λ and γ, and the entropy regularization parameter ε on four representative cross-scene datasets, A $\rightarrow$ U, R $\rightarrow$ W, W $\rightarrow$ U, and U $\rightarrow$ A. The shaded regions indicate stable operating intervals where TA-DOTCNet consistently achieves high accuracy, demonstrating robustness to hyperparameter selection. (a) Balancing coefficient λ. (b) Balancing coefficient γ. (c) Entropy regularization parameter ε.

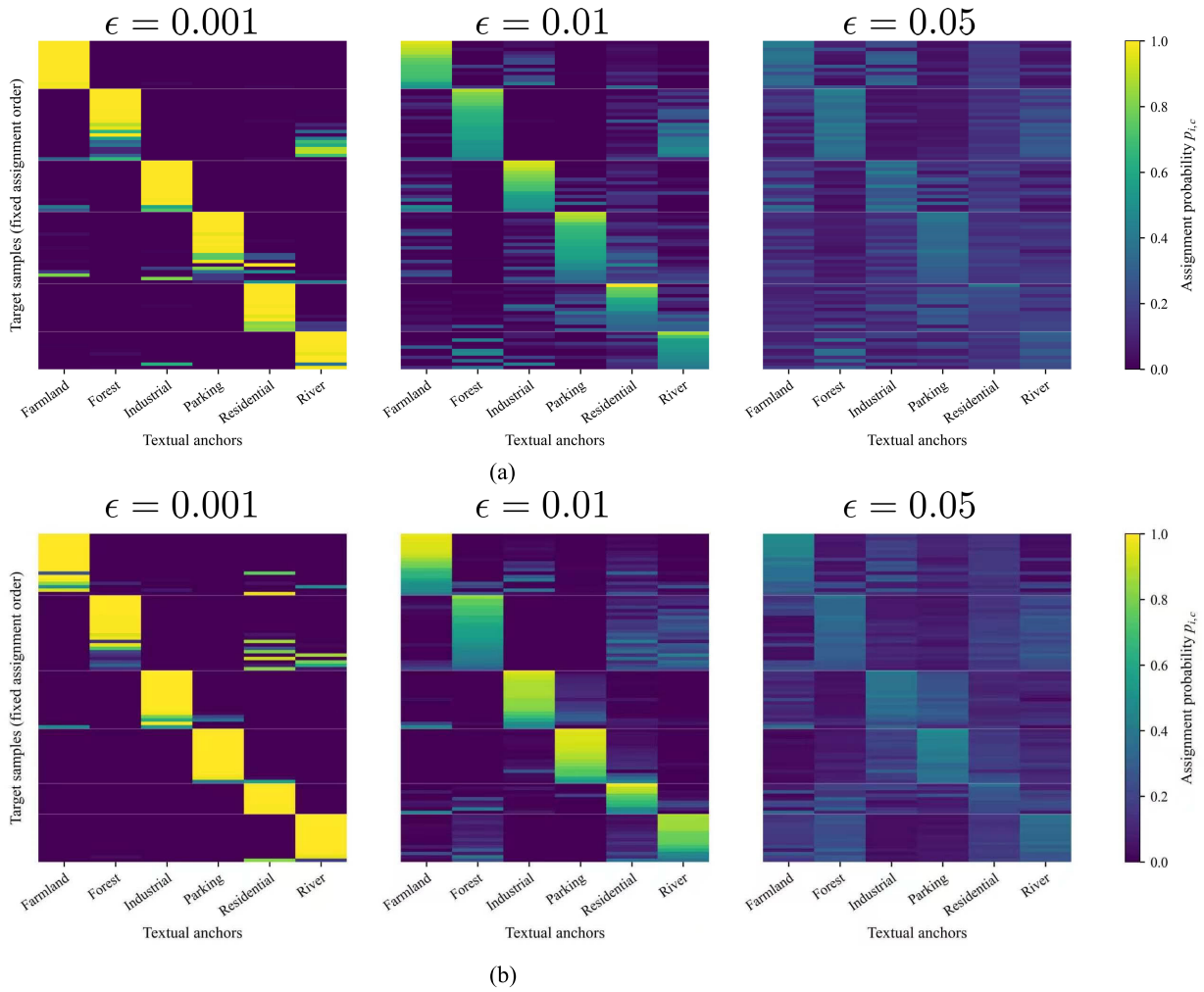


Fig. 6. Effect of entropy regularization intensity ϵ on the target distribution p on (a) U $\rightarrow$ R and (b) U $\rightarrow$ A datasets.

For each scene category, we initially generate five candidate descriptions and compute the averaged text embedding as the semantic representation. We further observe that the framework is relatively robust to the specific wording, i.e., replacing the averaged representation with a single representative description causes only minor performance changes. For simplicity and reproducibility, Table I not only reports the generated five candidate descriptions but also the representative descriptions adopted in the experiments. As shown in Section IV-F, the semantic embeddings we constructed exhibit higher separability compared to both raw image features and generic prompt templates.

2) Cross-Scene Alignment and Enhancement: To obtain semantically consistent feature representations across heterogeneous domains, TA-DOTCNet adopts a multimodal alignment and enhancement strategy built upon a shared

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TABLE I
LG CANDIDATE DESCRIPTIONS FOR EACH SCENE CATEGORY AND THE REPRESENTATIVE DESCRIPTION USED IN EXPERIMENTS

Category	LLM-Generated Candidate Descriptions	Representative Description Used in Experiments
Farmland	• Aerial view of farmland with organized crop fields, clear furrows and irrigation channels. • Satellite image of cultivated farmland showing rectangular plots and planting rows. • Top-down view of farmland with tractors, silos and farm infrastructure visible. • A large agricultural landscape of farmland with patchwork crop patterns. • Farmland dominated by seasonal crops with visible irrigation systems and drainage.	*Aerial view of farmland with organized crop fields, clear furrows and irrigation channels.*
Forest	• Aerial view of dense forest dominated by tall trees and a continuous canopy. • Satellite image of forest with mixed tree species, shrubs, and understory vegetation. • Forest area showing irregular canopy texture, clear tree crowns and shadows. • High-resolution overhead image of forest with winding trails and clearings. • Forest dominated by woody vegetation, exhibiting high biomass and biodiversity.	*Aerial view of dense forest dominated by tall trees and a continuous canopy.*
Industrial	• Aerial view of industrial area with large factory buildings, warehouses, and chimneys. • Satellite image of industrial zone with rectangular roof structures and parking lots. • Top-down image of industrial showing heavy infrastructure, shipping yards and storage tanks. • Industrial complex with roads, loading bays and container stacks visible from above. • High-resolution view of industrial with factories, warehouses and adjacent railway spurs.	*Aerial view of industrial area with large factory buildings, warehouses, and chimneys.*
Parking	• An overhead image of large parking lots with many parked vehicles and lanes. • Satellite photo of parking area showing organized parking rows and vehicle shadows. • Aerial view of commercial parking lots near shopping areas and road access. • Parking area with marked bays, drive aisles and adjacent access roads visible from above. • High-resolution top-down image of parking spaces and vehicle density patterns.	*Aerial view of commercial parking lots near shopping areas and road access.*
Residential	• Aerial view of residential neighborhood with houses, yards, and tree-lined streets. • Satellite photo of suburban residential area showing rooftops and garden plots. • Top-down image of residential blocks with streets, driveways and small gardens. • High-resolution overhead view of residential with homes, yards, swimming pools and local roads. • Residential area with mixed housing types, sidewalks and occasional green patches.	*Aerial view of residential neighborhood with houses, yards, and tree-lined streets.*
River	• Satellite image of river and riparian zone, showing winding water channels and banks. • Aerial view of river with adjacent floodplain vegetation and meandering course. • Top-down image showing river channels, oxbows and vegetated riverbanks. • High-resolution satellite photo of river with visible sediment plumes and islands. • River corridor with surrounding wetlands, aquatic vegetation and flood features.	*Aerial view of river with adjacent floodplain vegetation and meandering course.*

RemoteCLIP backbone. RemoteCLIP [48] is a VLM pretrained on large-scale RS image–text pairs, enabling it to learn land-cover semantic priors that surpass those of standard CLIP models trained solely on natural images [31], [49]. We employ RemoteCLIP as a shared image encoder for both source and target domains, forming a Siamese architecture to preliminarily extract semantically consistent image representations across diverse scene variations

$$\mathbf{z}_i^d = \mathcal{E}_{\text{img}}\left(\mathbf{x}_i^d\right), \quad d \in \{s, t\} \tag{2}$$

where $\mathbf{x}_i^d$ denotes the ith image sample from domain d, $\mathcal{E}_{\text{img}}(\cdot)$ represents the shared RemoteCLIP image encoder with frozen parameters, and $\mathbf{z}_i^d$ is the resulting image embedding. For textual embeddings, we build upon the previously introduced text prompt construction module and encode all category-specific prompt descriptions using the text encoder of RemoteCLIP. Specifically, each prompt is first tokenized by the RemoteCLIP tokenizer and then mapped into a shared vision–language

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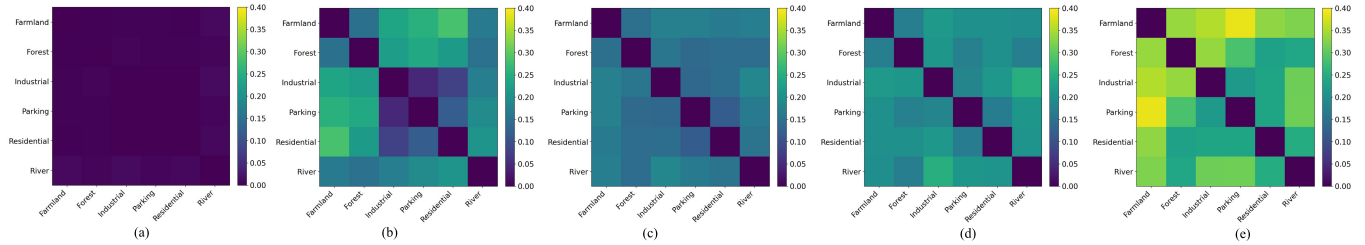


Fig. 7. Text prompt construction test. Normalized interclass semantic distance heatmaps for five settings. (a) Raw image features, (b) projected image embeddings, (c) single-noun prompts, (d) GT prompts, and (e) our constructed prompts. Brighter off-diagonal responses indicate larger interclass separation, showing that our prompts produce the most discriminative semantic structure.

embedding space through the frozen text encoder

$$\mathbf{t}_c = \mathcal{E}_{\text{text}}\left(\text{Tokenizer}\left(P_c\right)\right), \quad c = 1, \ldots, C \tag{3}$$

where P_c denotes the constructed prompt description for the cth scene category, Tokenizer$(\cdot)$ represents the tokenizer of RemoteCLIP, and $\mathcal{E}_{\text{text}}(\cdot)$ is the frozen RemoteCLIP text encoder. The resulting set of textual embeddings $\{\mathbf{t}_c\}_{c=1}^{C}$ forms a category-level semantic pool that is shared across domains. This semantic pool is exploited in two ways: 1) to construct prototype-driven counterparts for both source- and target-domain images (See Section III-B3) and 2) to serve as a semantic anchor in the Deep OT Cluster Head (See Section III-C).

After obtaining the frozen RemoteCLIP image embeddings, both source- and target-domain features are passed through a shared cross-scene adaptation layer $\mathcal{R}(\cdot)$. The module is implemented as a lightweight projection network consisting of normalization, two fully connected layers, a GRLU activation, and dropout layers, as illustrated in Fig. 2. The parameters of $\mathcal{R}(\cdot)$ are shared across the two domains in the Siamese framework, so that both branches are mapped into the same task-specific embedding space. In this way, $\mathcal{R}(\cdot)$ serves not only as a projection module, but also as a shared adaptation interface that connects frozen vision-language representations with subsequent semantic fusion and OT-based clustering. Source and target image embeddings are aligned at the representation level, ensuring that heterogeneous RS scenes are mapped into a unified semantic space. The refined image embeddings $\mathbf{r}_i^d = \mathcal{R}(\mathbf{z}_i^d)$ are concatenated with their corresponding prototype-driven textual counterparts to form joint multimodal representations, which are then fed into a classifier optimized with the cross-entropy loss $\mathcal{L}_{\text{CE}}$ to enhance scene-level discriminability

$$\mathbf{h}_i^d = \mathbf{r}_i^d \oplus \tilde{\mathbf{t}}_i^d, \quad d \in \{s, t\} \tag{4}$$

$$\hat{\mathbf{y}}_i^d = \mathcal{C}\left(\mathbf{h}_i^d\right) \tag{5}$$

where $\oplus$ denotes the concatenation operation, and $\tilde{\mathbf{t}}_i^d$ denotes its corresponding prototype-driven textual counterpart. The detailed construction of $\tilde{\mathbf{t}}_i^d$ is described in Section III-B3. $\mathcal{C}(\cdot)$ denotes the classifier trained under the cross-entropy loss $\mathcal{L}_{\text{CE}}$

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N_s}\sum_{i=1}^{N_s}\sum_{c=1}^{C}\mathbb{I}\left(y_i^s = c\right)\log p_{i,c}^s$$

$$-\frac{\gamma}{N_t}\sum_{j=1}^{N_t}\sum_{c=1}^{C}\mathbb{I}\left(\hat{y}_j^t = c\right)\log p_{j,c}^t. \tag{6}$$

N_s and N_t denote the numbers of samples in the source and target domains, respectively, and C is the total number of scene categories. $p_{i,c}^s$ and $p_{j,c}^t$ represent the predicted probabilities of class c for the ith source-domain sample and the jth target-domain sample, respectively. $\mathbb{I}(\cdot)$ is the indicator function, which equals 1 if the condition is satisfied and 0 otherwise. γ is the balancing coefficient controlling the contribution between source and target. Beyond supervised classification, TA-DOTCNet further enhances multimodal consistency through the deep OT cluster head, which will be introduced in Section III-C.

3) Prototype-Driven Counterpart Construction and Fusion: To explicitly incorporate category-level semantic information into image representations, we introduce a prototype-driven counterpart construction and fusion mechanism. This module dynamically constructs a textual counterpart for each image based on the relationship between image prototypes and the category-level semantic pool.

Specifically, we maintain a global image prototype that aggregates semantic information from all previously observed samples. At the beginning of training, the global prototype is initialized using the average embedding of the first mini-batch. As training proceeds, the global prototype is updated by incorporating the embeddings of the current mini-batch into the historical set and computing their cumulative mean

$$\mathbf{p}_{\text{global}}^d = \frac{1}{N_{\text{his}}^d + B}\left(N_{\text{his}}^d\,\mathbf{p}_{\text{global}}^d + \sum_{i=1}^{B}\mathbf{r}_i^d\right), \quad d \in \{s, t\} \tag{7}$$

where $\mathbf{r}_i$ denotes the refined image embedding after the cross-scene adaptive layers, B is the batch size, and N_{his} represents the total number of samples accumulated before the current mini-batch.

Given the updated global prototype, we then construct a sample-specific prototype for each image by integrating the global prototype with the corresponding image embedding through an exponential moving average

$$\mathbf{p}_i^d = \beta\,\mathbf{p}_{\text{global}}^d + (1-\beta)\,\mathbf{r}_i^d, \quad d \in \{s, t\} \tag{8}$$

which serves as the final prototype representation for the ith image. $\beta \in [0, 1)$ is the momentum coefficient controlling the update rate. For simplicity, it is fixed to 0.5, which provides an equal balance. This design enables each sample-level

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prototype to simultaneously encode long-term global semantic information accumulated across batches and instance-specific characteristics of the current image, thereby providing a stable yet adaptive reference for subsequent prototype-driven weighting and fusion.

Given the semantic pool of textual embeddings $\{\mathbf{t}_c\}_{c=1}^{C}$ introduced in Section III-B2, we compute the cosine similarity between the image-specific prototype and each textual embedding on both source and target domains

$$s_c^d = \frac{\left(\mathbf{p}^d\right)^\top \mathbf{t}_c}{\left\|\mathbf{p}^d\right\|_2 \ \|\mathbf{t}_c\|_2}, \quad d \in \{s,t\}, \quad c = 1,\ldots,C. \tag{9}$$

These similarity scores are normalized using a softmax function to obtain prototype-aware semantic coefficients

$$\alpha_c^d = \frac{\exp\left(s_c^d\right)}{\sum_{c'=1}^{C}\exp\left(s_{c'}^d\right)}, \quad d \in \{s,t\}. \tag{10}$$

Using these coefficients, we construct a prototype-driven textual counterpart by performing a weighted aggregation over the semantic pool

$$\tilde{\mathbf{t}}^d = \sum_{c=1}^{C} \alpha_c^d \mathbf{t}_c, \quad d \in \{s,t\}. \tag{11}$$

The resulting $\tilde{\mathbf{t}}^d$ captures category-level semantic information that is most relevant to the current image prototype, serving as an adaptive textual counterpart shared by samples within the batch.

Finally, as introduced in Section III-B2, the prototype-driven textual counterpart is concatenated with the refined image embedding to form a joint multimodal representation, which is subsequently fed into the classifier for scene recognition.

C. Deep OT Clustering

To further enforce semantic consistency between visual representations and category-level textual semantics, we introduce a deep OT cluster head. This module operates on the refined image embeddings produced by the cross-scene adaptive layers and leverages the textual semantic pool as fixed anchors. Notably, the entire cluster head is parameter-free and serves as a distributional regularizer during training. Fig. 3 shows the schematic illustration.

1) Semantic Distribution Refinement via OT: Given the refined image embeddings $\{\mathbf{r}_i^d\}_{i=1}^{N_d}$ from domain $d \in \{s,t\}$ and the textual semantic pool $\{\mathbf{t}_c\}_{c=1}^{C}$, we first compute a cost matrix using CLIPScore-based similarity. Each entry measures the similarity between an image embedding and a textual anchor

$$\mathbf{S}_{i,c}^d = \frac{\left(\mathbf{r}_i^d\right)^\top \mathbf{t}_c}{\left\|\mathbf{r}_i^d\right\|_2 \ \|\mathbf{t}_c\|_2}, \quad d \in \{s,t\} \tag{12}$$

where rows correspond to image samples and columns correspond to textual anchors. The similarity matrix $\mathbf{S}^d$ is then converted into a source distribution by row-wise normalization

$$q_{i,c}^d = \frac{\exp\left(1+\mathbf{S}_{i,c}^d\right)}{\sum_{c'=1}^{C}\exp\left(1+\mathbf{S}_{i,c'}^d\right)}, \quad d \in \{s,t\} \tag{13}$$

which represents the initial probability distribution of assigning image $\mathbf{r}_i^d$ to textual anchor $\mathbf{t}_c$.

EOT provides a principled way to align two distributions while preserving global mass consistency, and can be efficiently solved by the classical Sinkhorn-style iterative scaling procedure [43], [44]. In recent years, EOT has been widely adopted in machine learning as a flexible tool for distribution matching and structure-aware alignment across heterogeneous spaces [50], [51]. To obtain a more compact and semantically discriminative target distribution, we model the alignment between image embeddings and textual anchors as an EOT problem. Specifically, we solve for a transport plan $\mathbf{T}^d \in \mathbb{R}^{N_d \times C}$ by minimizing

$$\mathbf{T}^d = \arg\min_{\mathbf{T}\in\mathcal{U}} \left\langle \mathbf{T}, \mathbf{C}^d \right\rangle + \varepsilon \sum_{i,c} \mathbf{T}_{i,c} \log \mathbf{T}_{i,c}$$
$$\text{s.t. } \mathbf{T}^d \mathbf{1} = \boldsymbol{\mu}, \quad \left(\mathbf{T}^d\right)^\top \mathbf{1} = \boldsymbol{\nu}, \quad d \in \{s,t\} \tag{14}$$

where $\mathbf{C}^d = 1 - \mathbf{S}^d$ is the cost matrix, ε controls the entropy (and thus sparsity) of the transport plan, and $\mathcal{U}$ denotes the set of admissible transport matrices with prescribed marginals. $\boldsymbol{\mu} \in \mathbb{R}^{N_d}$ and $\boldsymbol{\nu} \in \mathbb{R}^{C}$ are the marginal distributions over image embeddings and textual anchors, respectively. In practice, we adopt fixed uniform marginals. The image-side marginal is defined as $\boldsymbol{\mu} = (1/N_d)1_{N_d}$, and the text-anchor-side marginal is defined as $\boldsymbol{\nu} = (1/C)1_C$.

The OT matrix $\mathbf{T}^d$ represents a joint probability distribution between image embeddings and textual anchors. By normalizing each row with respect to its marginal, we obtain a compact target assignment distribution

$$p_{i,c}^d = \frac{\mathbf{T}_{i,c}^d}{\sum_{c'=1}^{C}\mathbf{T}_{i,c'}^d}, \quad d \in \{s,t\} \tag{15}$$

whose shape can be adjusted by the entropy regularization coefficient ε, yielding either sharper or smoother assignment profiles.

The text-anchored clustering objective is then defined by minimizing the KL divergence between the source distribution q^d and the target distribution p^d

$$\mathcal{L}_{\mathrm{KL}}^{\mathrm{text}} = \sum_{d\in\{s,t\}} \sum_{i=1}^{N_d} \sum_{c=1}^{C} p_{i,c}^d \log \frac{p_{i,c}^d}{q_{i,c}^d}. \tag{16}$$

This loss encourages the image-to-text assignment distributions to evolve from similarity-driven soft assignments toward more compact, transport-optimized semantic alignments.

In addition to text-anchored alignment, we further impose a distributional regularization in the image embedding space. Instead of directly clustering images against textual anchors, we construct an image-anchored assignment matrix whose columns are formed by text-guided nearest image embeddings. Specifically, for each textual anchor $\mathbf{t}_c$, we retrieve its nearest image embedding from the union of refined embeddings (across both domains) according to cosine similarity

$$\mathbf{n}_c = \mathbf{r}_{j_c^*}, \quad j_c^* = \arg\max_j \frac{\mathbf{r}_j^\top \mathbf{t}_c}{\left\|\mathbf{r}_j\right\|_2 \ \|\mathbf{t}_c\|_2}, \quad c = 1,\ldots,C \tag{17}$$

where $\mathbf{n}_c$ serves as the image-space counterpart (anchor) associated with the cth textual semantics.

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Given an image embedding $\mathbf{r}_i$, we compute its similarity to each text-guided image anchor $\mathbf{n}_c$

$$\mathbf{S}^{\text{img}}_{i,c} = \frac{\mathbf{r}_i^\top \mathbf{n}_c}{\|\mathbf{r}_i\|_2 \ \|\mathbf{n}_c\|_2}, \quad c = 1, \ldots, C \tag{18}$$

and obtain a similarity-based source distribution by row-wise normalization

$$q^{\text{img}}_{i,c} = \frac{\exp\left(1 + \mathbf{S}^{\text{img}}_{i,c}\right)}{\sum_{c'=1}^{C} \exp\left(1 + \mathbf{S}^{\text{img}}_{i,c'}\right)}. \tag{19}$$

To derive a compact target distribution in the image space, we further refine $\mathbf{S}^{\text{img}}$ via an EOT problem analogous to (14), where the cost matrix is defined as $\mathbf{C}^{\text{img}} = 1 - \mathbf{S}^{\text{img}}$ and the same uniform marginal principle is also used. Solving this problem yields the transport plan $\mathbf{T}^{\text{img}}$, from which the target distribution is obtained by row-wise normalization

$$p^{\text{img}}_{i,c} = \frac{\mathbf{T}^{\text{img}}_{i,c}}{\sum_{c'=1}^{C} \mathbf{T}^{\text{img}}_{i,c'}}. \tag{20}$$

A KL divergence loss is applied to encourage consistent clustering structure across domains

$$\mathcal{L}^{\text{img}}_{\text{KL}} = \sum_{i} \sum_{c=1}^{C} p^{\text{img}}_{i,c} \log \frac{p^{\text{img}}_{i,c}}{q^{\text{img}}_{i,c}} \tag{21}$$

where q^{img} and p^{img} denote the similarity-based and compact target distributions in the image space, respectively.

The final clustering objective of the Deep OT cluster head is defined as follows:

$$\mathcal{L}_{\text{OTCluster}} = \mathcal{L}^{\text{text}}_{\text{KL}} + \mathcal{L}^{\text{img}}_{\text{KL}}. \tag{22}$$

By jointly optimizing text-anchored and image-space distributional alignment without introducing additional learnable parameters, the proposed cluster head effectively enhances semantic coherence and domain robustness of the learned representations.

2) Solving the Text-Anchored OT Problem: The proposed text-anchored EOT objective in (14) can be efficiently solved through its Lagrangian dual. For clarity, we drop the superscript d when no ambiguity arises and consider one domain at a time.

We start from the entropy-regularized primal problem

$$\begin{aligned} &\min_{\mathbf{T} \geq 0} \sum_{i=1}^{N_d} \sum_{c=1}^{C} \mathbf{T}_{i,c} \, \mathbf{C}_{i,c} + \varepsilon \sum_{i=1}^{N_d} \sum_{c=1}^{C} \mathbf{T}_{i,c} \log \mathbf{T}_{i,c} \\ &\text{s.t. } \mathbf{T}\mathbf{1} = \boldsymbol{\mu}, \quad \mathbf{T}^\top \mathbf{1} = \boldsymbol{\nu} \end{aligned} \tag{23}$$

where $\mathbf{C} = 1 - \mathbf{S}$ and $\boldsymbol{\mu}, \boldsymbol{\nu}$ are the prescribed marginals following fixed uniform distribution. Introducing dual variables $\mathbf{a} \in \mathbb{R}^{N_d}$ and $\mathbf{b} \in \mathbb{R}^{C}$ for the marginal constraints, the (augmented) Lagrangian is given by the following equation:

$$\begin{aligned} \mathcal{L}(\mathbf{T}, \mathbf{a}, \mathbf{b}) = &\sum_{i,c} \mathbf{T}_{i,c} \mathbf{C}_{i,c} + \varepsilon \sum_{i,c} \mathbf{T}_{i,c} \log \mathbf{T}_{i,c} \\ &+ \mathbf{a}^\top (\boldsymbol{\mu} - \mathbf{T}\mathbf{1}) + \mathbf{b}^\top \left(\boldsymbol{\nu} - \mathbf{T}^\top \mathbf{1}\right). \end{aligned} \tag{24}$$

Minimizing $\mathcal{L}$ w.r.t. $\mathbf{T}$ yields a closed-form stationary condition. Taking partial derivatives and setting them to zero leads to the following equation:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{T}_{i,c}} = \mathbf{C}_{i,c} + \varepsilon \left(1 + \log \mathbf{T}_{i,c}\right) - \mathbf{a}_i - \mathbf{b}_c = 0 \tag{25}$$

which implies

$$\mathbf{T}_{i,c} = \exp\left(\frac{\mathbf{a}_i + \mathbf{b}_c - \mathbf{C}_{i,c}}{\varepsilon} - 1\right). \tag{26}$$

Absorbing the constant factor $\exp(-1)$ into the dual scalings, the solution admits the standard factorized form.

Define the Gibbs kernel

$$\mathbf{K} = \exp\left(-\frac{\mathbf{C}}{\varepsilon}\right) \tag{27}$$

and re-parameterize the dual variables by the following equation:

$$\mathbf{u}_i = \exp\left(\frac{\mathbf{a}_i}{\varepsilon}\right), \quad \mathbf{v}_c = \exp\left(\frac{\mathbf{b}_c}{\varepsilon}\right). \tag{28}$$

Then the OT plan can be written compactly as follows:

$$\mathbf{T} = \operatorname{diag}(\mathbf{u}) \, \mathbf{K} \operatorname{diag}(\mathbf{v}) \tag{29}$$

where $\mathbf{u} \in \mathbb{R}^{N_d}$ and $\mathbf{v} \in \mathbb{R}^{C}$ are dual-induced scaling vectors that enforce the marginal constraints. To solve for $\mathbf{u}$ and $\mathbf{v}$, we enforce the constraints $\mathbf{T}\mathbf{1} = \boldsymbol{\mu}$ and $(\mathbf{T})^\top \mathbf{1} = \boldsymbol{\nu}$, yielding the coupled system

$$\mathbf{u} \odot (\mathbf{K}\mathbf{v}) = \boldsymbol{\mu}, \quad \mathbf{v} \odot \left((\mathbf{K})^\top \mathbf{u}\right) = \boldsymbol{\nu} \tag{30}$$

where $\odot$ denotes element-wise multiplication. This system is efficiently solved by alternating updates

$$\mathbf{u} \leftarrow \boldsymbol{\mu} \oslash (\mathbf{K}\mathbf{v}), \quad \mathbf{v} \leftarrow \boldsymbol{\nu} \oslash \left((\mathbf{K})^\top \mathbf{u}\right) \tag{31}$$

where $\oslash$ denotes element-wise division.

To improve numerical stability when ε is small (leading to sharper, more compact assignments), we implement the updates in the log-domain. Let $\mathbf{f} = \varepsilon \log \mathbf{u}$ and $\mathbf{g} = \varepsilon \log \mathbf{v}$ denote the dual potentials. Using $\mathbf{K} = \exp(-\mathbf{C}/\varepsilon)$ and the log-sum-exp operator, the updates can be equivalently written as follows:

$$\begin{cases} \mathbf{f} \leftarrow \varepsilon \log \boldsymbol{\mu} - \varepsilon \log \left(\displaystyle\sum_{c=1}^{C} \exp\left(\frac{\mathbf{g}_c - \mathbf{C}_{:,c}}{\varepsilon}\right) \right) \\ \mathbf{g} \leftarrow \varepsilon \log \boldsymbol{\nu} - \varepsilon \log \left(\displaystyle\sum_{i=1}^{N_d} \exp\left(\frac{\mathbf{f}_i - \mathbf{C}_{i,:}}{\varepsilon}\right) \right) \end{cases} \tag{32}$$

where $\mathbf{C}_{:,c}$ and $\mathbf{C}_{i,:}$ denote the cth column and ith row of $\mathbf{C}$, respectively. After convergence, $\mathbf{u}$ and $\mathbf{v}$ are recovered by $\mathbf{u} = \exp(\mathbf{f}/\varepsilon)$ and $\mathbf{v} = \exp(\mathbf{g}/\varepsilon)$. The converged transport plan $\mathbf{T}$ constitutes a joint distribution over images and textual anchors. Finally, we obtain the compact target assignment distribution by row-wise normalization via (15). Notably, ε provides explicit control over the compactness of the resulting assignments. Specifically, ε acts as a temperature-like parameter in the Gibbs kernel $\mathbf{K}$. A smaller ε makes $\mathbf{T}$ more sensitive to the semantic cost, concentrating transport mass on low-cost image–text pairs and producing a sharper row-normalized target distribution p. A larger ε weakens cost differences

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Algorithm 1 Forward Procedure of the DOTC Module

Input: Refined image embeddings $\{\mathbf{r}_i^d\}_{i=1}^{N_d}$ from domain $d \in \{s,t\}$, textual semantic anchors $\{\mathbf{t}_c\}_{c=1}^{C}$, entropy regularization coefficient ϵ
Output: Text-anchored clustering loss $\mathcal{L}_{\mathrm{OTCluster}}$
foreach *domain* $d \in \{s,t\}$ **do**
  Step 1: Compute image-text similarity matrix $\mathbf{S}_{i,c}^d$ via Eq. (12);
  Step 2: Construct similarity-based source distribution $q_{i,c}^d$ by row-wise normalization via Eq. (13);
  Step 3: Build cost matrix

$$\mathbf{C}^d = 1 - \mathbf{S}^d;$$

  Step 4: Construct the Gibbs kernel $\mathbf{K}^d$ via Eq. (27);
  Step 5: Solve the entropy-regularized OT problem with Sinkhorn iterations to obtain transport plan $\mathbf{T}^d$:
    Step 5.1: Initialize scaling vectors and uniform marginals

$$\mathbf{u}^d = \mathbf{1}, \qquad \mathbf{v}^d = \mathbf{1};$$

$$\boldsymbol{\mu} = (1/N_d)1_{N_d}, \qquad \boldsymbol{\nu} = (1/C)1_C;$$

    for $l = 1$ **to** L_{Max} **do**
      Step 5.2: Update row scaling $\mathbf{u}^d$ via Eq. (31);
      Step 5.3: Update column scaling $\mathbf{v}^d$ via Eq. (31);
    Step 5.4: Recover the transport plan $\mathbf{T}^d$ via Eq. (29);
  Step 6: Normalize each row of $\mathbf{T}^d$ to obtain compact target distribution $p_{i,c}^d$ via Eq. (15);
  Step 7: Compute text-anchored KL loss $\mathcal{L}_{\mathrm{KL}}^{\mathrm{text}}$ via Eq. (16);
Step 8: Construct image-space anchors $\{\mathbf{n}_c\}_{c=1}^{C}$ by retrieving the nearest refined image embedding to each textual anchor via Eq. (17);
Step 9: Compute image-image similarity matrix $\mathbf{S}_{i,c}^{\mathrm{img}}$ via Eq. (18);
Step 10: Construct similarity-based image-space source distribution $q_{i,c}^{\mathrm{img}}$ via Eq. (19);
Step 11: Build image-space cost matrix

$$\mathbf{C}^{\mathrm{img}} = 1 - \mathbf{S}^{\mathrm{img}};$$

Step 12: Solve the entropy-regularized OT problem with Sinkhorn iterations to obtain $\mathbf{T}^{\mathrm{img}}$, following Step 5;
Step 13: Normalize each row of $\mathbf{T}^{\mathrm{img}}$ to obtain $p_{i,c}^{\mathrm{img}}$ via Eq. (20);
Step 14: Compute image-space KL loss $\mathcal{L}_{\mathrm{KL}}^{\mathrm{img}}$ via Eq. (21);
Step 15: Obtain the DOTC loss $\mathcal{L}_{\mathrm{OTCluster}}$ via Eq. (22);
return $\mathcal{L}_{\mathrm{OTCluster}}$;

and increases the entropy of $\mathbf{T}$, leading to a smoother and more diffused p. Therefore, ε controls the trade-off between compact semantic assignment and smooth balanced transport. Importantly, this entire procedure is parameter-free and serves as a principled distributional refinement step guided by the text anchors.

To improve implementation transparency, we provide a dedicated pseudocode description of the DOTC forward procedure in Algorithm 1, which explicitly summarizes the execution order of similarity computation, Sinkhorn-based transport refinement, target distribution construction, and final KL-based loss.

IV. Experimental Study

In this section, we will validate the effectiveness and advanced performance of our proposed TA-DOTCNet on benchmark datasets. Numerical results and visualizations will be reported. Cross-scene tasks encompass not only *Source* → *Target* domain but also *Target* → *Source* domain.

A. Description of Datasets, Evaluation, and Experimental Configuration

1) Datasets Details: To comprehensively evaluate the proposed cross-scene RSSC framework, we conduct experiments on four widely used aerial image datasets, namely AID, UCM, WHU, and RSSCN7. These datasets are collected under diverse imaging conditions and exhibit notable variations in spatial resolution, scene composition, and acquisition environments, making them suitable benchmarks for this area, as shown in Fig. 4.

1) *AID [52]:* This dataset contains 10 000 images distributed across 30 aerial scene classes. Each class comprises 220–420 samples with a size of 600 × 600 pixels. The pixel resolution varies significantly from 0.5 to 8 m. Due to the multisource collection from different imaging sensors, AID exhibits strong intraclass diversity and scale variations, posing a challenge for feature invariance.
2) *UCM [53]:* The UC Merced Land Use dataset consists of 2100 high-resolution aerial images covering 21 land-use categories, with exactly 100 images per class. Each image measures 256 × 256 pixels with a fine spatial resolution of 0.3 m. While the scenes are visually coherent, the limited sample size requires models to have high data efficiency to avoid overfitting during adaptation.
3) *WHU-RS19 [54]:* Collected from Google Earth, this dataset includes 1005 images categorized into 19 structural classes (e.g., airports, viaducts). It contains approximately 50 samples per class with a resolution of 600 × 600 pixels. The dataset is characterized by complex spatial structures and varying illumination, introducing nonnegligible domain shifts when aligned with other datasets.
4) *RSSCN7 [55]:* This dataset comprises 2800 images from seven typical scene categories, with 400 images per class and a size of 400 × 400 pixels. A unique feature of RSSCN7 is that its samples are acquired across different seasons and weather conditions. This introduces significant appearance variability (e.g., color and texture shifts) rather than purely structural differences, making it an ideal benchmark for testing robustness against environmental changes.

To ensure a fair and consistent evaluation under a closed-set domain adaptation setting, following the common practice in existing research [17], [29], we carefully align the label spaces of all datasets. Specifically, six common scene categories shared across the four datasets are selected, including *Residential*, *Farmland*, *Forest*, *Industrial*, *Parking*, and *River*. All images belonging to other categories are excluded. This

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preprocessing strategy guarantees that the source and target domains share an identical set of semantic classes while still preserving substantial domain shifts in visual appearance and data distribution.

In all experiments, one dataset is treated as the labeled source domain, while another dataset is used as the unlabeled target domain. The model is trained using labeled samples from the source domain and unlabeled samples from the target domain. Based on the four benchmark datasets, a total of 12 cross-scene classification subtasks can be constructed for evaluation. Specifically, AID, UCM, WHU-RS19, and RSSCN7 are abbreviated as A, U, W, and R, respectively. Accordingly, the cross-scene tasks include A $\rightarrow$ U, A $\rightarrow$ W, A $\rightarrow$ R, U $\rightarrow$ W, U $\rightarrow$ A, U $\rightarrow$ R, W $\rightarrow$ U, W $\rightarrow$ A, W $\rightarrow$ R, R $\rightarrow$ U, R $\rightarrow$ W, and R $\rightarrow$ A. This experimental protocol strictly follows the unsupervised domain adaptation paradigm and allows for an objective assessment of cross-domain generalization performance under realistic RS scenarios.

2) Evaluation and Experimental Configuration: This section describes the evaluation protocol and experimental configuration used in our experiments, including evaluation metrics, implementation details, and baseline methods, to ensure a fair and reproducible comparison.

a) Evaluation metrics: In the experiment, we primarily utilize overall accuracy (OA) to quantitatively assess the effectiveness of the proposed method. OA represents the ratio of correctly predicted samples to the total number of test samples in the target domain and is calculated as follows:

$$\mathrm{OA} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(\hat{y}_i = y_i) \tag{33}$$

where N denotes the total number of target test samples, y_i is the ground-truth label, $\hat{y}_i$ is the predicted label, and $\mathbb{I}(\cdot)$ is the indicator function which equals 1 if the condition is true and 0 otherwise. OA serves as the standard benchmark for evaluating cross-scene generalization in RSSC.

b) Implementation details: The proposed framework is built upon a RemoteCLIP backbone with a ViT-B/32 architecture, initialized using publicly available pretrained weights [48]. Both the image encoder and text encoder are kept frozen throughout training to preserve their general-purpose semantic representations. A lightweight projection network is introduced to map CLIP image features into a task-specific embedding space, as shown in Fig. 2. For generating textual descriptions used for each scene category provided in Table I, we utilize the "Qwen-Plus" API.

The model is optimized using stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 5×10^{-4}. The learning rate is set to 0.02. The batch size is fixed to 16 for both source and target domains, and all models are trained for 200 epochs. All experiments are implemented using the PyTorch framework and conducted on a single NVIDIA RTX 4090 GPU. In addition, three hyperparameters are empirically determined. The first is the balancing coefficient λ, which controls the relative contribution between the classification loss and the clustering loss. The second is the balancing coefficient γ in the classification loss between source and target. The last is the entropy regularization parameter ε, which governs the entropy of the OT plan and thus determines the sparsity of the resulting assignment. Specifically, a larger ε leads to smoother target distributions, while a smaller value yields more compact and sharper assignments. The sensitivity of these hyperparameters will be analyzed in Section IV-E.

c) Overview of comparison algorithms: To comprehensively evaluate the proposed method, we compare TA-DOTCNet with several representative baselines including conventional supervised classifiers ResNet [56] and ViT [57]; classical domain adaptation methods JAN [58] and MRAN [59]; state-of-the-art methods specifically designed for RS cross-scene AMRAN [29], BSCA [17], SCSN [60], APDM [63], PUMCL [61], and SFMDA [65]; and recent prompt-based adaptation methods PDA [62] and PADA-Net [64].

B. Ablation Study

Table II reports the ablation results of TA-DOTCNet on 12 cross-scene target-domain tasks. Purely image-based baselines, including ResNet and ViT, achieve limited average accuracies of 78.03% and 86.90%, respectively, while source–target joint training brings only moderate improvements, indicating that visual supervision alone is insufficient under severe cross-scene shift. Replacing them with frozen RemoteCLIP features already yields a strong baseline of 85.66%, confirming the benefit of RS vision-language pretraining. On this basis, adding the shared adapter and cross-entropy loss improves the average performance to 90.87% and 92.35%, showing that task-specific adaptation remains necessary even with a strong pretrained backbone. Further introducing prototype-driven textual counterparts raises the average accuracy to 93.30%, demonstrating the effectiveness of external category-level semantic guidance. Finally, incorporating the proposed OTCluster achieves the best result of 95.62%, with a particularly clear gain on the most challenging transfer U $\rightarrow$ R, improving from 78.54% to 85.71%. Overall, these results show that the improvement of TA-DOTCNet is not solely due to the pretrained backbone, but comes from the complementary effects of semantic-guided counterpart construction and OT-based clustering refinement.

C. Comparison With State-of-the-Art Models on Target Domain

Table III reports the target-domain OA on 12 cross-scene tasks. Conventional supervised baselines show limited transferability under cross-scene shift, with ResNet and ViT achieving average accuracies of 78.03% and 86.90%, respectively. Classical domain adaptation methods, such as JAN and MRAN, improve performance to 80.74% and 84.48%, but their gains remain moderate, suggesting that visual-level distribution alignment alone is insufficient to ensure high-level semantic consistency. More recent cross-scene methods further improve target-domain generalization, e.g., BSCA reaches 90.97%, while APDM and SFMDA achieve 88.73% and 90.18%, respectively. Prompt-based methods also show strong competitiveness, with PADA-Net obtaining 95.05% on average.

In contrast, TA-DOTCNet achieves the best average accuracy of 95.62% and outperforms existing methods on most

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TABLE II

ABLATION STUDY (OA% ↑) OF TA-DOTCNET ON 12 CROSS-SCENE TARGET-DOMAIN TASKS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

Setting	Img	Text	Backbone	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R→U	R→W	R→A	Avg
Src-only train	✓		ResNet-50	71.20	66.44	56.50	75.14	93.11	71.25	74.86	98.42	76.88	80.71	86.71	85.18	78.03
	✓		ViT-B/32	90.80	92.68	80.00	77.32	95.38	78.54	75.36	99.20	83.70	84.11	95.20	90.54	86.90
Src+Tgt joint train	✓		ResNet-50	82.00	74.44	61.77	83.39	91.39	70.16	85.18	98.40	76.88	81.96	91.60	85.75	81.91
	✓		ViT-B/32	95.60	94.43	85.42	87.50	97.07	84.32	87.50	99.60	85.57	93.57	97.72	94.37	91.85
No train	✓		RemoteCLIP	95.60	94.93	69.79	82.32	94.93	69.79	82.32	95.60	69.79	82.32	95.60	94.93	85.66
+Adapter+CELoss (Src-only train)	✓		RemoteCLIP	96.00	92.68	76.98	87.68	97.24	73.23	94.64	99.20	81.09	98.21	98.80	94.71	90.87
+Adapter+CELoss (Src+Tgt joint train)	✓		RemoteCLIP	98.00	95.05	78.23	87.32	97.35	78.18	98.39	99.20	82.92	98.57	98.80	96.17	92.35
+Adapter+Prototype-driven Counterpart+CELoss	✓	✓	RemoteCLIP	98.00	95.21	78.54	95.00	97.80	79.79	98.91	99.32	82.96	98.64	98.85	96.61	93.30
+Adapter+Prototype-driven Counterpart+CELoss+OTCluster	✓	✓	RemoteCLIP	**98.40**	**97.92**	**85.71**	**99.46**	**98.52**	**86.00**	**99.11**	**99.60**	**86.21**	**99.11**	**99.37**	**98.02**	**95.62**

tasks. The gains are especially evident on several challenging transfers, including U → R, A → U, and W → U. These results indicate that TA-DOTCNet not only improves overall target-domain accuracy, but also yields more stable performance across datasets with different levels of scene discrepancy. The advantage mainly comes from the semantic-guided design, where category-level textual anchors help align image features toward shared semantic concepts rather than domain-specific appearance patterns.

D. Comparison With State-of-the-Art Models on Source Domain

Table IV reports the source-domain classification accuracy across 12 cross-scene settings. Here, ResNet and ViT are marked with † and included only as reference baselines for source-domain-only supervised fitting. Since they are trained and evaluated purely on the source domain, they provide a practically near-saturated reference level for examining whether cross-scene adaptation methods can preserve source-domain discriminability while improving target-domain generalization.

Most adaptation methods, including JAN, MRAN, and AMRAN, maintain relatively high source-domain performance, with average accuracies generally around 96%–97%. This suggests that domain adaptation does not severely damage source-domain representations, although noticeable fluctuations still appear on some transfers. More recent approaches, such as BSCA, APDM, and PDA, show comparable source-domain performance, achieving average accuracies of 96.71%, 96.74%, and 97.08%, respectively. While these methods are primarily designed to improve target-domain transfer, their results suggest that they are able to retain a reasonable level of source-domain discrimination, albeit with noticeable variations on more challenging datasets. In comparison, TA-DOTCNet achieves the highest average source-domain accuracy of 98.13%, remaining very close to the source-only reference baselines while outperforming other adaptation methods on most tasks. Combined with the target-domain results in Table III, this demonstrates that TA-DOTCNet improves cross-scene transfer without sacrificing source-domain semantic discriminability. This behavior highlights the benefit of semantic-guided alignment, which encourages category-consistent representations instead of overfitting to domain-specific visual statistics.

E. Sensitivity Study

To evaluate the robustness of TA-DOTCNet, we analyze three key hyperparameters: the loss balancing coefficient λ, the target-domain classification loss ratio γ, and the entropy regularization parameter ε. Experiments are conducted on four representative cross-scene tasks, namely A → U, R → W, W → U, and U → A, as shown in Fig. 5. We adopt a grid-search strategy by fixing two parameters and sweeping the remaining one. Specifically, λ is searched in $\{0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 1.75, 2.00\}$, γ in $\{0.0025, 0.0050, 0.0075, 0.0100, 0.0125, 0.0150, 0.0175\}$, and ε in $\{0.005, 0.01, 0.05, 0.1, 0.5, 1\}$. As shown in Fig. 5, TA-DOTCNet maintains consistently strong performance within broad practical intervals. The best results are generally obtained at $\lambda \in [1.0, 1.5]$, $\gamma \in [0.0075, 0.0125]$, and $\varepsilon \in [0.05, 0.5]$. In particular, W → U and R → W remain above 98.0% across wide ranges, while A → U stays above 98.5% under suitable ε.

Finally, we adopt a single hyperparameter configuration for all experiments for convenience: $\lambda = 1.25$, $\gamma = 0.01$, and $\varepsilon = 0.1$. These values fall within the stable regions shown in Fig. 5 and consistently yield strong performance on A → U, R → W, W → U, and U → A. Similar trends are also observed in other datasets. Although performance varies when the hyperparameters are swept across a wide range, the method maintains consistently strong results within a broad and practically meaningful interval.

To directly show how ϵ affects the transport-refined target distribution, we visualize p under different ϵ values as in Fig. 6. When ϵ is small, the assignment distribution is highly concentrated, indicating that the transport mass is mainly allocated to a few low-cost image–text pairs. As ϵ increases, the distribution gradually becomes smoother and more diffused.

F. Analysis and Visualization

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TABLE III

OA% ↑ BETWEEN OUR METHOD AND COMPARISON ALGORITHMS ON 12 CROSS-SCENE DATASETS ON TARGET DOMAIN, THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN ITALIC

Method	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R→U	R→W	R→A	Avg
ResNet [56] (Src. Only)	71.20	66.44	56.50	75.14	93.11	71.25	74.86	98.42	76.88	80.71	86.71	85.18	78.03
ViT [57] (Src. Only)	90.80	92.68	80.00	77.32	95.38	78.54	75.36	99.20	83.70	84.11	95.20	90.54	86.90
JAN [58]	76.58	69.19	61.89	81.86	92.29	73.58	82.86	98.10	78.13	81.57	88.61	84.28	80.74
MRAN [59]	78.32	82.99	64.78	81.71	94.06	79.16	84.71	98.73	78.96	83.06	93.35	93.87	84.48
AMRAN [29]	89.56	84.80	69.02	85.20	94.82	79.50	88.16	*99.68*	81.00	87.56	94.59	94.48	87.36
BSCA [17]	90.51	90.42	77.31	91.04	97.23	83.60	90.78	99.57	85.63	90.58	98.29	96.72	90.97
SCSN [60]	74.40	68.36	68.80	63.93	94.82	71.15	61.96	98.80	73.23	91.60	79.11	90.26	78.04
PUMCL [61]	77.55	75.92	61.06	77.12	94.08	79.06	84.50	98.68	78.68	85.25	89.97	85.44	82.28
PDA [62]	*97.15*	95.07	79.33	87.33	*97.43*	81.08	83.67	99.37	78.25	86.17	97.47	95.99	89.86
APDM [63]	93.20	93.02	74.01	79.11	95.83	84.70	82.14	99.20	85.16	86.61	94.40	97.41	88.73
PADA-Net [64]	96.84	*95.90*	**88.04**	*95.83*	96.80	**87.79**	*97.00*	**100.00**	**87.33**	*98.33*	*98.73*	*97.97*	*95.05*
SFMDA [65]	90.40	92.68	70.67	85.71	97.01	85.20	90.71	98.80	85.33	91.07	97.20	97.36	90.18
TA-DOTCNet (Ours)	**98.40**	**97.92**	*85.71*	**99.46**	**98.52**	***86.00***	**99.11**	99.60	*86.21*	**99.11**	**99.37**	**98.02**	**95.62**

1) Text Prompt Construction Test: To evaluate how different text prompt designs affect semantic discriminability, we compare five types of representations in Fig. 7. The normalized interclass distance heatmaps show that the original image features in Fig. 7(a) exhibit weak class discrimination, while the projected image embeddings in Fig. 7(b) reveal only limited structural improvement. For text prompts, single nouns (SNs) in Fig. 7(c) provide only coarse semantic cues, and GTs in Fig. 7(d) improve alignment but still suffer from insufficient category specificity. In contrast, our constructed prompts in Fig. 7(e) produce a much clearer block-diagonal pattern with larger interclass margins and reduced cross-category confusion, indicating stronger category separability and a more reliable semantic basis for subsequent cross-scene alignment and clustering.

In addition to the semantic distance heatmaps in Fig. 7, Table V provides a quantitative comparison of different prompt strategies, including SNs, GTs, LG descriptions, and the representative description used (RU) in our experiments. Both LG and RU consistently outperform SN and GT on most tasks, indicating that richer category-level descriptions provide more effective semantic guidance than minimal or generic prompts. Moreover, LG achieves the best average accuracy of 94.50%, while RU reaches a closely matched 94.34%, suggesting that the proposed semantic pool is relatively robust to the exact wording as long as the descriptions capture characteristic land-cover components, spatial arrangements, and functional attributes.

2) IIR Analysis: To quantitatively evaluate the semantic separability of the learned representations under cross-scene shifts, we define the interclass/intraclass ratio (IIR), which measures the relative strength of interclass dispersion against intraclass compactness

$$\mathcal{IIR} = \frac{\frac{1}{C(C-1)} \sum_{c=1}^{C} \sum_{\substack{c'=1 \\ c' \neq c}}^{C} \|\mathbf{z}_c - \mathbf{z}_{c'}\|_2}{\frac{1}{C} \sum_{c=1}^{C} \mathrm{Var}(\mathbf{z}_c)} \tag{34}$$

where C denotes the number of scene categories and $\mathbf{z}_c$ represents the embedding of category c. A larger IIR indicates that samples from different categories are more separable while samples within the same category remain more clustered, reflecting a representation space that is both discriminative and category-consistent.

Table VI reports the IIR scores on three representative cross-scene tasks (U → W, U → R, and U → A). The ResNet baseline yields consistently low IIR values; Avg is 1.08, indicating limited class separability and weak semantic consistency when transferring across scenes. Using RemoteCLIP features improves IIR to an average of 1.60, suggesting that vision–language pretraining brings a modest gain in forming more structured and semantically meaningful clusters, yet the improvement remains limited under substantial scene variations. In contrast, TA-DOTCNet achieves a substantially higher IIR across all tasks, Avg is 15.99, demonstrating a pronounced enhancement in representation separability and intraclass compactness simultaneously. Notably, the gain is particularly significant on U → W and U → A, where TA-DOTCNet improves IIR from 2.18 to 17.91 and from 1.74 to 25.39, respectively. These results indicate that our semantic-guided framework effectively organizes cross-scene features into category-consistent clusters, alleviating the tendency of image-only alignment to mix categories or fragment the same category across domains. Overall, the IIR analysis provides quantitative evidence that TA-DOTCNet learns a more

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TABLE IV

OA% ↑ BETWEEN OUR METHOD AND COMPARISON ALGORITHMS ON 12 CROSS-SCENE DATASETS ON SOURCE DOMAIN. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN ITALIC. † INDICATES THE METHOD IS NOT INCLUDED IN THE COMPETITIVE COMPARISON AND IS REPORTED ONLY AS A REFERENCE BASELINE

Method	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R→U	R→W	R→A	Avg
ResNet† [56]	100.00	100.00	100.00	98.48	98.48	98.48	99.32	99.32	99.32	97.50	97.50	97.50	98.82
ViT† [57]	97.86	97.86	97.86	96.97	96.97	96.97	98.87	98.87	98.87	87.29	97.29	97.29	97.75
JAN [58]	96.43	97.86	97.86	*98.48*	96.97	96.97	98.65	97.97	98.87	93.54	92.71	94.17	96.71
MRAN [59]	99.29	**99.98**	95.57	93.94	97.89	95.45	98.87	99.10	99.32	93.96	93.56	94.58	96.79
AMRAN [29]	99.29	98.57	98.57	96.97	96.97	95.45	98.98	99.32	99.01	92.13	94.19	93.86	96.94
BSCA [17]	97.86	*99.29*	*99.29*	92.42	*98.48*	95.45	98.87	99.10	99.32	94.17	92.29	93.96	96.71
PUMCL [61]	91.50	85.00	84.50	75.80	94.54	90.90	91.23	94.20	96.14	77.50	71.25	70.00	85.21
PDA [62]	96.82	98.93	97.31	97.20	98.40	*98.40*	99.05	99.32	95.43	*95.76*	94.58	93.78	*97.08*
APDM [63]	*99.29*	93.57	96.43	98.47	96.97	96.97	*99.10*	99.32	98.65	95.21	93.75	93.12	96.74
PADA-Net [64]	97.86	97.86	90.00	98.43	98.00	96.97	98.55	*99.32*	99.55	95.04	*94.83*	95.08	96.79
SFMDA [65]	96.25	97.92	98.21	97.20	98.40	96.67	97.01	98.14	*99.55*	92.13	89.58	*95.54*	96.38
TA-DOTCNet (Ours)	**99.29**	98.57	**99.29**	**98.48**	**98.48**	**98.48**	**99.10**	**99.55**	**99.55**	**95.83**	**95.42**	**95.62**	**98.13**

TABLE V

COMPARISON OF DIFFERENT TEXT PROMPT STRATEGIES ON SIX DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**. SN: SINGLE NOUN, GT: GENERIC TEMPLATE, LG: LLM-GENERATED, RU: REPRESENTATIVE USED

Dataset	SN	GT	LG	RU
U→W	98.20	98.20	**99.20**	98.40
U→A	97.41	97.80	97.64	**97.92**
U→R	85.20	85.67	**85.78**	85.71
W→U	99.28	99.23	99.46	**99.46**
W→A	98.16	98.21	**98.93**	98.52
W→R	84.92	85.80	86.00	**86.00**
Avg	93.86	94.15	**94.50**	94.34

TABLE VI

IIR ↑ OF FEATURE REPRESENTATIONS ON THREE CROSS-SCENE TASKS. HIGHER VALUES INDICATE STRONGER INTERCLASS SEPARABILITY RELATIVE TO INTRACLASS COMPACTNESS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND THE SECOND-BEST IN ITALIC

Method	U→W	U→R	U→A	Avg
ResNet [56]	1.46	0.71	1.08	1.08
RemoteCLIP [48]	*2.18*	*0.89*	*1.74*	*1.60*
TA-DOTCNet (Ours)	**17.91**	**4.67**	**25.39**	**15.99**

discriminative and semantically consistent embedding space for robust cross-scene classification.

3) Effect of the Cross-Scene Adapter: Table VII compares three adaptation strategies: no adaptation, separate adaptation layers for the two domains, and the proposed shared adaptation layer. Without adaptation, the model already achieves 90.23% average accuracy, reflecting the strong baseline provided by RemoteCLIP. Using separate adaptation layers yields only a limited improvement to 91.29%, suggesting that independently adapting the two domains does not effectively preserve a unified semantic space. In contrast, the proposed shared adapter achieves the best average accuracy of 95.62%, with clear gains on several challenging transfers. These results show that the cross-scene adaptation layer is necessary, and that parameter sharing is important for learning a common task-specific embedding space.

4) Feature Visualization: To qualitatively analyze the effect of semantic-guided alignment, we visualize the learned target-domain feature distributions using t-SNE [66] on three representative cross-scene tasks, i.e., U → A, R → A, and W → A, as shown in Fig. 8. For the ResNet baseline, features from different scene categories exhibit noticeable overlap and ambiguous boundaries, particularly for visually correlated classes such as Residential and Industrial, indicating that purely appearance-driven representations struggle to maintain category-level consistency under domain shift. In contrast, TA-DOTCNet yields well-separated clusters with significantly reduced intraclass variance across all evaluated tasks. Samples belonging to the same category form compact and coherent groups, while different categories are clearly separated in the embedding space. This improvement suggests that semantic-guided alignment encourages the model to organize features

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TABLE VII
STUDY OF DIFFERENT CROSS-SCENE ADAPTATION STRATEGIES ON 12 TARGET-DOMAIN CROSS-SCENE TASKS

Adaptation Strategy	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R→U	R→W	R→A	Avg
No adaptation	97.60	94.37	74.90	86.79	96.11	76.09	86.25	98.80	78.39	98.21	99.20	96.06	90.23
Separate adaptation	96.40	91.89	76.98	94.82	97.75	77.34	95.89	98.80	80.36	89.46	98.00	97.75	91.29
Shared adaptation (Ours)	98.40	97.92	85.71	99.46	98.52	86.00	99.11	99.60	86.21	99.11	99.37	98.02	95.62

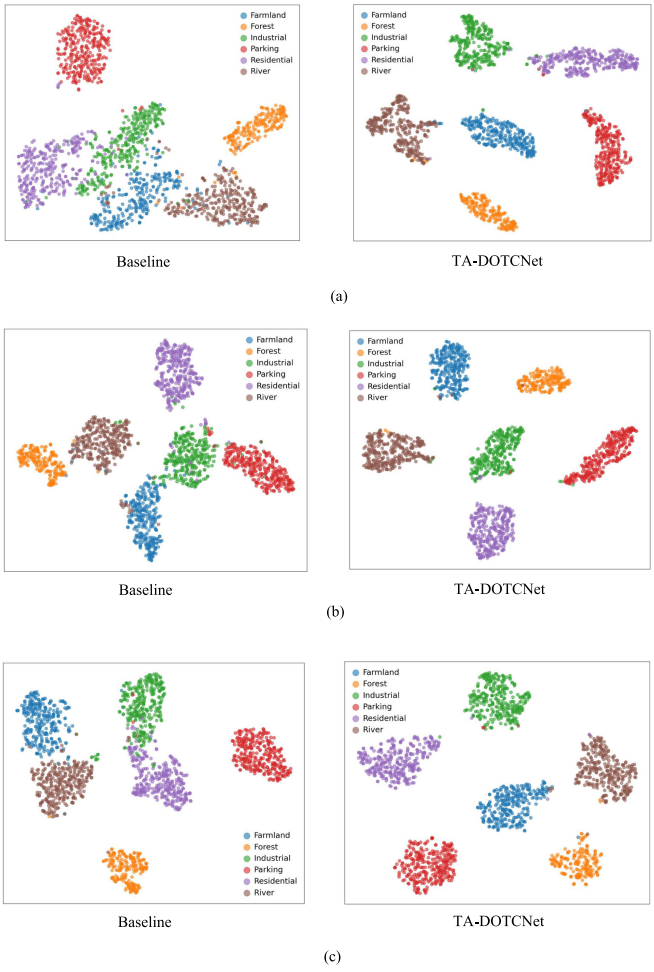


Fig. 8. t-SNE visualization of target-domain feature representations on three cross-scene tasks (a) U → A, (b) R → A, and (c) W → A. (Left) Features extracted by the ResNet baseline trained without semantic guidance. (Right) Features learned by TA-DOTCNet.

according to category-consistent concepts rather than domain-specific appearance statistics.

5) Computational Efficiency: Compared with the frozen RemoteCLIP backbone, the extra cost of TA-DOTCNet mainly comes from the shared adaptation layer, prototype-driven semantic aggregation, and the DOTC module. Let B, C, D, and L denote the batch size, number of categories, embedding dimension, and Sinkhorn iterations, respectively. The adaptation layer has complexity approximately $\mathcal{O}(BD^2)$, while the prototype-driven text fusion requires about $\mathcal{O}(BCD)$. For DOTC, each transport refinement involves cost matrix construction with complexity $\mathcal{O}(BCD)$ and Sinkhorn updates with complexity $\mathcal{O}(LBC)$. Since the text encoder is frozen, the semantic pool is small, and DOTC is parameter-free, the additional overhead remains moderate in practice. Therefore, the overall computational burden is still dominated by the shared RemoteCLIP backbone. Table VIII compares the computational efficiency of TA-DOTCNet with several recent methods. TA-DOTCNet achieves a favorable trade-off between accuracy and efficiency. Although its parameter size is comparable to PDA, it is substantially more efficient, with markedly fewer FLOPs (5.90 G versus over 115 G) and faster training and inference speed than PADA-Net. Compared with lighter models such as PUMCL and SFMDA, TA-DOTCNet is less efficient in terms of raw speed and model size, but still remains computationally practical while delivering stronger cross-scene performance.

TABLE VIII
COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT METHODS. TRAINING SPEED IS MEASURED IN BATCHES/S, INFERENCE SPEED IN SAMPLES/S

Method	Params (M)	Train. Speed	Infer. Speed	FLOPs (G)
PDA [62]	152.83	3.43	320.24	115.67
PADA-Net [64]	90.60	0.54	12.03	21.08
SFMDA [65]	24.43	8.27	478.16	3.64
PUMCL [61]	26.15	10.08	1511.86	4.11
TA-DOTCNet	152.34	7.33	141.71	5.90

V. CONCLUDING REMARKS

This article presents TA-DOTCNet, a semantic-guided cross-scene classification framework that explicitly incorporates category-level textual knowledge into RS scene understanding. By introducing a text-anchored external guidance strategy, TA-DOTCNet leverages pretrained VLMs to provide stable semantic anchors, enabling cross-scene representations to move beyond image-only alignment and reduce domain-specific bias. In addition, a parameter-free DOTC module is proposed to refine similarity-based assignments into compact, semantically consistent target distributions. This distributional regularization enhances both text-anchored semantic alignment and image-space structural consistency without introducing extra learnable parameters. Extensive experiments on multiple benchmark datasets demonstrate that TA-DOTCNet consistently outperforms SOTA methods, yielding more discriminative and domain-invariant representations. Despite the encouraging results, TA-DOTCNet is still limited to closed-set cross-scene classification and relies on predefined LG category-level semantic descriptions. In addition, although

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the proposed OT-based clustering head introduces no extra learnable parameters, its iterative refinement may become less efficient in larger-scale scenarios. Future work will explore more flexible semantic construction, more scalable transport optimization, and broader cross-scene settings.

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